

TEZPUR UNIVERSITY

Test V, Autumn 2014

C0 504 Natural Language Processing

Marks : 25

Time : 30 minutes

The figures in the right-hand margin indicate marks for the individual question.

1. Suggest a suitable tag-set and perform Part of Speech (PoS) tagging of the following sentence- 2 + 2

Kamal Kumar clearly said he would shop in the morning for Diwali.

2. Give an example of a PoS tagging rule and a situation where it can be applied. 3
3. Suggest a context free grammar for imperative sentences in English. Demonstrate with an example how it can be used to describe the structure of a sentence. 2 + 2
4. Give two examples of different types of structural ambiguities in English? 3
5. What is the use of *feature structures* of sentence components? What would be the result of unification of the feature structures $[NUM \ SL]$ and $[PERS \ 2]$? 2 + 1
6. What is the use of *probabilistic context free grammars*? How can the probabilities of the grammar rules be estimated? 3
7. Represent the following sentence using First Order Predicate Calculus- 2
Tezpur University students use bicycles.

8. Identify lexical semantic relationships among the following words- 3

bit (verb), chair, beat (music), too, furniture, beat (assault), two, beat (result of an encounter), defeat, bit (noun)